



Data Engineer

Tech Interview Titans

Interview Questions & Answers

Experience: 2-5 Year

Round -3 (Technical + Managerial)

1. Explain your end-to-end data pipeline from source to dashboard.

Answer

In my last project, data was coming from multiple sources like **transactional databases, APIs, and flat files**.

First, I used **Kafka** to ingest real-time events and **batch ingestion** using scheduled jobs for historical data.

Then the raw data landed in a **data lake (S3/ADLS)**.

From there:

- I cleaned and transformed data using **Spark (PySpark)**
- Applied business rules
- Stored curated data in **Delta / Parquet format**
- Loaded final tables into a **Data Warehouse** for reporting

BI tools like **Power BI / Tableau** were connected to the warehouse for dashboards.

I also handled **data validation, logging, and failure alerts**.

What interviewer checks

- Can you explain clearly?
- Do you understand *why* each tool is used?



2. What challenges did you face in production pipelines?

Answer

The biggest challenges were:

- **Late-arriving data**
- **Schema changes**
- **Job failures due to data quality issues**

For late data, I used **watermarking and reprocessing logic**.

For schema changes, I implemented **schema evolution** and validations before processing.

For failures, I added:

- Retry mechanisms
- Proper logging
- Alerts using email / monitoring tools

I learned that **monitoring is as important as writing code**.

3. How do you optimize Spark jobs?

Answer

I focus on **both data and Spark configuration**.

Key things I do:

- Avoid `collect()` on large datasets
- Use **partitioning & bucketing**
- Cache only reusable DataFrames
- Use **broadcast joins** for small tables
- Tune executor memory and cores

Before optimization, I always check:

- Data size
- Skew
- Execution plan

4. How do you handle data quality?

Answer

Data quality is handled at **multiple levels**:

- Null checks



- Duplicate detection
- Range validation
- Referential integrity checks

If data fails validation:

- It goes to a **quarantine table**
- Alerts are triggered
- Pipeline continues without breaking

This avoids blocking business dashboards.

5. Difference between Data Lake and Data Warehouse?

Answer

- **Data Lake** stores raw, semi-structured, and structured data
- **Data Warehouse** stores cleaned, structured, business-ready data

Data Lake = storage first, schema later

Data Warehouse = schema first, analytics ready

We use both because they solve different problems.

6. How do you handle incremental data loads?

Answer

I mostly use:

- **Watermark columns**
- **Last updated timestamp**
- **CDC (Change Data Capture)**

Only new or updated records are processed, which improves performance and reduces cost.

7. Have you worked with real-time data?

Answer

Yes. We used **Kafka** for real-time ingestion.

Flow:

- Producer sends events
- Spark Structured Streaming consumes data
- Applies transformations



- Writes to storage with checkpoints

Checkpointing helps in **fault tolerance and exactly-once processing**.

8. Managerial: How do you handle tight deadlines?

Answer

First, I break the work into **small deliverables**.

Then I:

- Identify critical paths
- Communicate clearly with stakeholders
- Deliver MVP first
- Optimize later

If something is risky, I inform early instead of hiding issues.

9. Managerial: What if your pipeline fails at night?

Answer

I ensure:

- Automated alerts
- Clear logs
- Runbooks for common issues

If it's critical:

- I analyze logs
- Restart jobs if safe
- Inform stakeholders with ETA

Post-incident, I always do **root cause analysis** to avoid repeat failures.

10. Managerial: How do you work with business teams?

Answer

I don't jump into coding immediately.

First, I understand:

- Business problem
- Metrics definition
- Data expectations



I explain technical limitations in **simple language**, not jargon.
This avoids rework and builds trust.

11. What makes a good Data Engineer?

Answer

A good Data Engineer:

- Writes clean, scalable code
- Thinks about data quality
- Understands business impact
- Communicates well
- Takes ownership of pipelines

It's not just about tools, it's about **responsibility**.

12. Why should we hire you?

Answer

I bring:

- Strong fundamentals
- Real production experience
- Problem-solving mindset
- Ownership attitude

I focus on **building reliable systems**, not just completing tasks.

